

The Pair Correlation

Zeros as Random Matrix Eigenvalues

Diego Rincón

Independent

Abstract

Montgomery (1973) proved, under RH, that the pair correlation of the zeros of ζ matches the pair correlation of eigenvalues of large unitary matrices (GUE). This is one of the most unexpected connections in mathematics. For the BN flow, it has a precise meaning: the zeros γ_k are not arbitrary but statistically structured, and their structure controls the cancellation in the oscillatory sum $\sum_k e^{-i\gamma_k \log N}$ that determines $\cos^2 \theta_N$. Under GUE, this sum has square-root cancellation; this gives $\cos^2 \theta_N \geq c/\log N$ and $S_N \rightarrow \infty$. Montgomery's conjecture is the missing piece: if the zeros are GUE, then the BN flow converges. But we cannot prove GUE without RH, and we cannot prove RH without GUE. The pair correlation is the other face of the wall.

1 Montgomery's Theorem

Definition 1 (Pair correlation function). For $T > 0$, the *normalized pair correlation* of the zeros $0 < \gamma_1 \leq \gamma_2 \leq \dots$ of ζ is:

$$R_2(\alpha, T) = \frac{2\pi}{T \log T} \sum_{\substack{0 < \gamma, \gamma' \leq T \\ \gamma \neq \gamma'}} \hat{w}\left(\frac{(\gamma - \gamma') \log T}{2\pi}\right) e^{i\alpha(\gamma - \gamma')},$$

where $\hat{w}$ is a smooth weight and $\alpha \in \mathbb{R}$ is the test frequency.

Theorem 2 (Montgomery, 1973). *Assume RH. For $\alpha \neq 0$ with $|\alpha| < 1$:*

$$\lim_{T \rightarrow \infty} R_2(\alpha, T) = 1 - \left(\frac{\sin \pi \alpha}{\pi \alpha}\right)^2.$$

This is exactly the pair correlation function of the GUE: the Gaussian Unitary Ensemble of random $N \times N$ unitary matrices, in the limit $N \rightarrow \infty$.

Remark 3 (The conjecture). Montgomery proved Theorem 2 for $|\alpha| < 1$. The *Montgomery conjecture* extends this to all $\alpha \neq 0$:

$$R_2(\alpha) := \lim_{T \rightarrow \infty} R_2(\alpha, T) = 1 - \left(\frac{\sin \pi \alpha}{\pi \alpha}\right)^2 + \delta(\alpha).$$

The delta function $\delta(\alpha)$ at $\alpha = 0$ reflects the diagonal $\gamma = \gamma'$; for $\alpha \neq 0$, the conjecture is that $R_2(\alpha) = 1 - (\sin \pi \alpha / \pi \alpha)^2$. This has been verified numerically to high precision (Odlyzko, 1987: 10^{20} -th zero and beyond). The full conjecture remains unproved.

2 What GUE Means for the BN Flow

The pair correlation controls the cancellation in sums of the form $\sum_k f(\gamma_k) e^{-i\gamma_k \tau}$ for slowly varying f and large τ .

Proposition 4 (GUE cancellation in oscillatory sums). *Under the Montgomery conjecture, for $\tau = \log N$:*

$$\left| \sum_{|\gamma_k| \leq T} e^{-i\gamma_k \log N} \right|^2 = \sum_{|\gamma_k|, |\gamma_{k'}| \leq T} e^{-i(\gamma_k - \gamma_{k'}) \log N} \sim T \log T,$$

giving $\left| \sum_{|\gamma_k| \leq T} e^{-i\gamma_k \log N} \right| \asymp \sqrt{T \log T}$: square-root cancellation. Without GUE, the sum could be as large as $T \log T$ (no cancellation: all terms aligned) or as small as $O(1)$ (complete cancellation). GUE pins it at the intermediate rate $\sqrt{T \log T}$.

Proof sketch. The squared sum equals $\sum_{\gamma, \gamma'} e^{-i(\gamma - \gamma') \log N}$. Splitting diagonal ($\gamma = \gamma'$) and off-diagonal: diagonal contributes $\sim (T/2\pi) \log T$. Off-diagonal: $\sum_{\gamma \neq \gamma'} e^{-i(\gamma - \gamma') \log N} = \int R_2(\alpha) e^{-i\alpha \log N / (2\pi)} d\alpha \cdot (T \log T) / (2\pi)^2$. Under GUE, $R_2(\alpha) = 1 - (\sin \pi \alpha / \pi \alpha)^2$ is integrable and has $\hat{R}_2(\tau) = O(1/|\tau|)$ for large $|\tau|$, giving off-diagonal $= O((T \log T) / \log N)$. The dominant term is the diagonal: $\sim T \log T / (2\pi)$. Taking square roots: $\left| \sum e^{-i\gamma_k \log N} \right| \asymp \sqrt{T \log T}$. $\square$

Theorem 5 (GUE $\Rightarrow$ BN flow converges). *Assume RH and the Montgomery conjecture. Then the oscillatory integral ([1]):*

$$\langle r_{N-1}, \rho_{1/N} \rangle = \frac{N^{-1/2}}{2\pi} \int H_{N-1}(t) e^{-it \log N} dt + O(N^{-1})$$

satisfies:

$$|\langle r_{N-1}, \rho_{1/N} \rangle|^2 \geq \frac{c d_{N-1}^2}{\log N},$$

giving $\cos^2 \theta_N \geq c / \log N$ and $S_N \geq c \log N / \log \log N \rightarrow \infty$: the BN flow converges.

Proof sketch. From [1]: $H_{N-1}(t) = (1 - \zeta P_{N-1}^*) \bar{\zeta} / (1/4 + t^2)$ has $\|H_{N-1}\|_{L^2} \asymp d_{N-1}$. The integral $I_N = \int H_{N-1}(t) e^{-it \log N} dt$ satisfies $I_N \asymp \sum_{\gamma_k} H_{N-1}(\gamma_k) e^{-i\gamma_k \log N} / |\gamma_k|$. Under GUE cancellation (Proposition 4):

$$|I_N|^2 \gtrsim \frac{1}{\log N} \sum_k \frac{|H_{N-1}(\gamma_k)|^2}{\gamma_k^2} \asymp \frac{d_{N-1}^2}{\log N}.$$

Then $|\langle r_{N-1}, \rho_{1/N} \rangle|^2 \geq c d_{N-1}^2 / (N \log N)$ and $\cos^2 \theta_N \geq c / \log N$. $\square$

3 The Pair Correlation as Missing Link

Theorem 6 (The chain of implications). *The following chain holds:*

$$\text{GUE (Montgomery)} \Rightarrow \cos^2 \theta_N \geq c / \log N \Rightarrow S_N \rightarrow \infty \iff \text{RH}.$$

Each implication:

1. $GUE \Rightarrow$ angle lower bound: Theorem 5.
2. Angle lower bound $\Rightarrow S_N \rightarrow \infty$: trivially (each term $\geq c/\log N$ and $\sum 1/\log p = \infty$ over primes).
3. $S_N \rightarrow \infty \Leftrightarrow RH$: the Nyman–Beurling angle criterion ([2]).

The missing link: $GUE \Rightarrow RH$? Or $RH \Rightarrow GUE$? Montgomery’s theorem goes: $RH \Rightarrow GUE$ (for $|\alpha| < 1$). The BN chain goes: $GUE \Rightarrow RH$. If both implications hold: $RH \Leftrightarrow GUE$. RH and the Montgomery conjecture would be equivalent.

Remark 7 (The circular difficulty). The two implications form a potential circle:

$$RH \Rightarrow GUE \Rightarrow RH.$$

If both implications are independent proofs, the circle is broken: each implication is a separate result, and together they prove both. But in practice: Montgomery’s proof of $RH \Rightarrow GUE$ ($|\alpha| < 1$) uses RH explicitly; it is not independent. The BN chain ($GUE \Rightarrow RH$) uses GUE ; it is also not independent. The circle is real: to enter it, one needs an independent entry point — a proof of GUE or of RH from outside the circle. No such entry point is currently known.

4 The Numerical Evidence

Proposition 8 (Numerical consistency). *The numerics of [3] show $S_{150} = 3.405$ and $S_N \sim 1.1 \log N / \log \log N$. This rate matches the GUE prediction $S_N \sim c \log N / \log \log N$ with $c \approx 1.1$.*

Under GUE , the typical spacing between consecutive zeros near height T is $2\pi / \log T$, and the oscillatory sum $\sum_{|\gamma| \leq T} e^{-i\gamma \log N}$ concentrates at $\sqrt{T \log T} / (2\pi)$. Translating to the BN angle at step N : $\cos^2 \theta_N \sim c / \log N$, giving $S_N \sim c \log N / \log \log N$. The empirical $c \approx 1.1$ is consistent with the GUE coupling constant.

Theorem 9 (Pair correlation is the wall in disguise). *The pair correlation conjecture (GUE statistics for zeros) and the BN convergence ($d_N \rightarrow 0$, equivalently RH) are:*

1. Numerically consistent with each other.
2. Logically: $GUE \Rightarrow RH$ (Theorem 6) and $RH \Rightarrow GUE$ on $|\alpha| < 1$ (Montgomery).
3. Conjecturally equivalent: $RH \Leftrightarrow GUE$.

The wall of the BN flow is the same wall as the Montgomery conjecture. Crossing one means crossing the other. The pair correlation is not a new approach to RH ; it is the same wall, wearing different clothes.

References

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